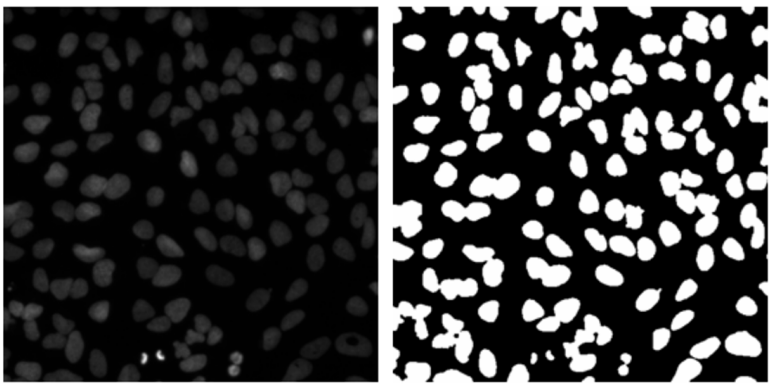
Tutorial Questions | Week 7

COSC2779 - Deep Learning

This tutorial is aimed at reviewing practical methodology in developing CNN. Please try the questions before you join the session. The question below has similar structure to the case study questions that you can expect in the end semester test.

1. You have been hired by a provider of pathology services in Victoria to solve one of their challenges in dealing with cell images: determining which parts of a microscope image corresponds to which individual cells. In deep learning, image segmentation is the process of assigning a label to every pixel in an image such that pixels with the same label share certain characteristics. In your case, you want to locate the cells and their boundaries in microscopic images. In other words, this is a classification task where each pixel of the target image is labeled as 0 (this pixel is not part of a cell) or 1 (this pixel is part of a cell).

   Here is an examples of an input image and the corresponding target image:

   The pathology company has collected 10,000 images from microscopes and gave them to you with annotations made by their doctors. The doctors have labeled each image as shown in the above figure, where all the pixels that belong to cells are labeled 1 and the other pixels that belong to the background are labelled 0. Images have been taken from three types of microscopes: A (5,000 images), B (2,500 images) and C (2,500 images). The doctors who hired you would like to use your algorithm on images from microscope C.

   (a) Explain how you would split this dataset into train, dev and test sets. Give the exact percentage split, and give reasons to your choices.

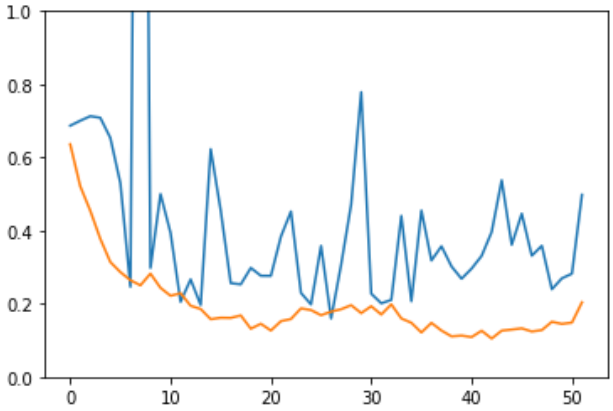
   (b) Can you augment this dataset? If yes, give only 3 distinct methods you would use. If no, explain why (give only 2 reasons).

   (c) The first step in building the system is to identify the performance measures, What performance measure would you select and why?

   (d) You have decided to use a CNN based approach to solve this problem, What type of base network architecture would you use? Why?

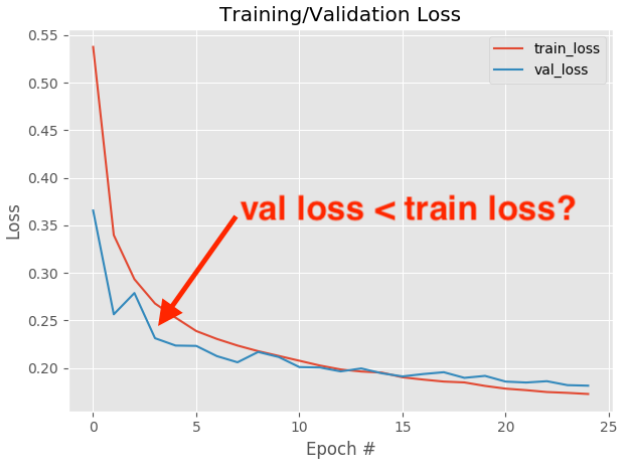
   (e) What will be a suitable cost function to train the model?

   (f) While training the model you observed the following loss curve:



What are the possible alteration you might try?

(g) You have now fixed the issues with your model and the best learning curve you obtained is as follows:



How would you explain, Why the validation loss lower than training loss? what would you do?